

SQL JOINS (E-Commerce Database)

```
CREATE DATABASE ecommerce_db;
```

```
USE ecommerce_db;
```

```
CREATE TABLE Categories (  
    category_id VARCHAR(10) PRIMARY KEY,  
    category_name VARCHAR(50)  
);
```

```
CREATE TABLE Suppliers (  
    supplier_id VARCHAR(10) PRIMARY KEY,  
    supplier_name VARCHAR(100),  
    city VARCHAR(50),  
    country VARCHAR(50)  
);
```

```
CREATE TABLE Products (  
    product_id VARCHAR(10) PRIMARY KEY,  
    product_name VARCHAR(100),  
    category_id VARCHAR(10),  
    supplier_id VARCHAR(10),  
    unit_price DECIMAL(10,2),  
    stock_quantity INT,
```

```
    FOREIGN KEY (category_id)  
        REFERENCES Categories(category_id),
```

```
FOREIGN KEY (supplier_id)
REFERENCES Suppliers(supplier_id)
);
```

```
CREATE TABLE Customers (
    customer_id VARCHAR(10) PRIMARY KEY,
    customer_name VARCHAR(100),
    gender VARCHAR(10),
    city VARCHAR(50),
    phone VARCHAR(20),
    email VARCHAR(100)
);
```

```
CREATE TABLE Employees (
    employee_id VARCHAR(10) PRIMARY KEY,
    employee_name VARCHAR(100),
    designation VARCHAR(50),
    city VARCHAR(50),
    salary DECIMAL(10,2)
);
```

```
CREATE TABLE Shippers (
    shipper_id VARCHAR(10) PRIMARY KEY,
    shipper_name VARCHAR(100),
    phone VARCHAR(20)
);
```

```
CREATE TABLE Orders (
```

order_id VARCHAR(10) PRIMARY KEY,
customer_id VARCHAR(10),
employee_id VARCHAR(10),
shipper_id VARCHAR(10),
order_date DATE,
order_status VARCHAR(30),

FOREIGN KEY (customer_id)
REFERENCES Customers(customer_id),

FOREIGN KEY (employee_id)
REFERENCES Employees(employee_id),

FOREIGN KEY (shipper_id)
REFERENCES Shippers(shipper_id)

);

CREATE TABLE Order_Items (
order_item_id VARCHAR(10) PRIMARY KEY,
order_id VARCHAR(10),
product_id VARCHAR(10),
quantity INT,
unit_price DECIMAL(10,2),

FOREIGN KEY (order_id)
REFERENCES Orders(order_id),

FOREIGN KEY (product_id)
REFERENCES Products(product_id)

);

INSERT INTO Categories VALUES

('CAT001','Electronics'),
('CAT002','Furniture'),
('CAT003','Clothing'),
('CAT004','Books'),
('CAT005','Home Appliances');

INSERT INTO Suppliers VALUES

('SUP001','ABC Electronics','Chennai','India'),
('SUP002','Home Decor Ltd','Bangalore','India'),
('SUP003','Fashion World','Mumbai','India'),
('SUP004','Book House','Delhi','India'),
('SUP005','Kitchen Solutions','Hyderabad','India');

INSERT INTO Products VALUES

('PRO001','Laptop','CAT001','SUP001',55000,50),
('PRO002','Smartphone','CAT001','SUP001',25000,100),
('PRO003','Office Chair','CAT002','SUP002',4500,80),
('PRO004','T-Shirt','CAT003','SUP003',800,200),
('PRO005','SQL Book','CAT004','SUP004',650,150),
('PRO006','Microwave Oven','CAT005','SUP005',12000,40);

INSERT INTO Customers VALUES

('CUS001','Rahul','Male','Chennai','9876543210','rahul@gmail.com'),
('CUS002','Priya','Female','Bangalore','9876543211','priya@gmail.com'),

```
('CUS003','John','Male','Hyderabad','9876543212','john@gmail.com'),  
('CUS004','Ayesha','Female','Delhi','9876543213','ayesha@gmail.com'),  
('CUS005','Kumar','Male','Coimbatore','9876543214','kumar@gmail.com');
```

INSERT INTO Employees VALUES

```
('EMP001','Arun','Sales Executive','Chennai',45000),  
('EMP002','Meena','Sales Executive','Bangalore',48000),  
('EMP003','David','Manager','Hyderabad',75000),  
('EMP004','Fatima','Warehouse Executive','Delhi',42000),  
('EMP005','Suresh','Supervisor','Chennai',60000);
```

INSERT INTO Shippers VALUES

```
('SHIP001','Blue Dart','90000000001'),  
('SHIP002','DTDC','90000000002'),  
('SHIP003','Delhivery','90000000003'),  
('SHIP004','India Post','90000000004'),  
('SHIP005','Ecom Express','90000000005');
```

INSERT INTO Orders VALUES

```
('ORD001','CUS001','EMP001','SHIP001','2026-07-01','Delivered'),  
('ORD002','CUS002','EMP002','SHIP002','2026-07-02','Shipped'),  
('ORD003','CUS003','EMP003','SHIP003','2026-07-03','Pending'),  
('ORD004','CUS004','EMP004','SHIP004','2026-07-04','Delivered'),  
('ORD005','CUS005','EMP005','SHIP005','2026-07-05','Processing');
```

INSERT INTO Order_Items VALUES

```
(
    'OI001','ORD001','PRO001',1,55000),
    ('OI002','ORD001','PRO004',2,800),
    ('OI003','ORD002','PRO002',1,25000),
    ('OI004','ORD003','PRO006',1,12000),
    ('OI005','ORD004','PRO003',2,4500),
    ('OI006','ORD005','PRO005',3,650);
```

```
SELECT * FROM Categories;
```

Result Grid		
Filter Rows:		
	category_id	category_name
▶	CAT001	Electronics
	CAT002	Furniture
	CAT003	Clothing
	CAT004	Books
	CAT005	Home Appliances
*	NULL	NULL

```
SELECT * FROM Suppliers;
```

Result Grid				
Filter Rows:				
	supplier_id	supplier_name	city	country
▶	SUP001	ABC Electronics	Chennai	India
	SUP002	Home Decor Ltd	Bangalore	India
	SUP003	Fashion World	Mumbai	India
	SUP004	Book House	Delhi	India
	SUP005	Kitchen Solutions	Hyderabad	India
*	NULL	NULL	NULL	NULL

SELECT * FROM Products;

Result Grid

Filter Rows:

Edit:

Export/Import:

	product_id	product_name	category_id	supplier_id	unit_price	stock_quantity
▶	PRO001	Laptop	CAT001	SUP001	55000.00	50
	PRO002	Smartphone	CAT001	SUP001	25000.00	100
	PRO003	Office Chair	CAT002	SUP002	4500.00	80
	PRO004	T-Shirt	CAT003	SUP003	800.00	200
	PRO005	SQL Book	CAT004	SUP004	650.00	150
	PRO006	Microwave Oven	CAT005	SUP005	12000.00	40
✱	NULL	NULL	NULL	NULL	NULL	NULL

SELECT * FROM Customers;

Result Grid

Filter Rows:

Edit:

Export/Import:

	customer_id	customer_name	gender	city	phone	email
▶	CUS001	Rahul	Male	Chennai	9876543210	rahul@gmail.com
	CUS002	Priya	Female	Bangalore	9876543211	priya@gmail.com
	CUS003	John	Male	Hyderabad	9876543212	john@gmail.com
	CUS004	Ayesha	Female	Delhi	9876543213	ayesha@gmail.com
	CUS005	Kumar	Male	Coimbatore	9876543214	kumar@gmail.com
✱	NULL	NULL	NULL	NULL	NULL	NULL

SELECT * FROM Employees;

Result Grid

Filter Rows:

Edit:

Export/Import:

	employee_id	employee_name	designation	city	salary
▶	EMP001	Arun	Sales Executive	Chennai	45000.00
	EMP002	Meena	Sales Executive	Bangalore	48000.00
	EMP003	David	Manager	Hyderabad	75000.00
	EMP004	Fatima	Warehouse Executive	Delhi	42000.00
	EMP005	Suresh	Supervisor	Chennai	60000.00
✱	NULL	NULL	NULL	NULL	NULL

SELECT * FROM Shippers;

	shipper_id	shipper_name	phone
▶	SHIP001	Blue Dart	9000000001
	SHIP002	DTDC	9000000002
	SHIP003	Delhivery	9000000003
	SHIP004	India Post	9000000004
	SHIP005	Ecom Express	9000000005
★	NULL	NULL	NULL

SELECT * FROM Orders;

	order_id	customer_id	employee_id	shipper_id	order_date	order_status
▶	ORD001	CUS001	EMP001	SHIP001	2026-07-01	Delivered
	ORD002	CUS002	EMP002	SHIP002	2026-07-02	Shipped
	ORD003	CUS003	EMP003	SHIP003	2026-07-03	Pending
	ORD004	CUS004	EMP004	SHIP004	2026-07-04	Delivered
	ORD005	CUS005	EMP005	SHIP005	2026-07-05	Processing
★	NULL	NULL	NULL	NULL	NULL	NULL

SELECT * FROM Order_Items;

	order_item_id	order_id	product_id	quantity	unit_price
▶	OI001	ORD001	PRO001	1	55000.00
	OI002	ORD001	PRO004	2	800.00
	OI003	ORD002	PRO002	1	25000.00
	OI004	ORD003	PRO006	1	12000.00
	OI005	ORD004	PRO003	2	4500.00
	OI006	ORD005	PRO005	3	650.00
★	NULL	NULL	NULL	NULL	NULL

Display all customers with their orders

```
SELECT
    c.customer_name,
    o.order_id,
    o.order_date,
    o.order_status
FROM Customers c
INNER JOIN Orders o
ON c.customer_id = o.customer_id;
```

OUTPUT

Result Grid

Filter Rows:

Export:

	customer_name	order_id	order_date	order_status
▶	Rahul	ORD001	2026-07-01	Delivered
	Priya	ORD002	2026-07-02	Shipped
	John	ORD003	2026-07-03	Pending
	Ayesha	ORD004	2026-07-04	Delivered
	Kumar	ORD005	2026-07-05	Processing

Display order details with customer names

```
SELECT
    o.order_id,
    c.customer_name,
    o.order_date,
    o.order_status
FROM Orders o
INNER JOIN Customers c
ON o.customer_id = c.customer_id;
```

OUTPUT

	order_id	customer_name	order_date	order_status
▶	ORD001	Rahul	2026-07-01	Delivered
	ORD002	Priya	2026-07-02	Shipped
	ORD003	John	2026-07-03	Pending
	ORD004	Ayesha	2026-07-04	Delivered
	ORD005	Kumar	2026-07-05	Processing

Show every product with its category

SELECT

p.product_name,

c.category_name

FROM Products p

INNER JOIN Categories c

ON p.category_id = c.category_id;

OUTPUT

	product_name	category_name
▶	Laptop	Electronics
	Smartphone	Electronics
	Office Chair	Furniture
	T-Shirt	Clothing
	SQL Book	Books
	Microwave Oven	Home Appliances

Display products with supplier names

SELECT

p.product_name,

```

        s.supplier_name
FROM Products p
INNER JOIN Suppliers s
ON p.supplier_id = s.supplier_id;

```

OUTPUT

Result Grid   Filter Rows:		
	product_name	supplier_name
▶	Laptop	ABC Electronics
	Smartphone	ABC Electronics
	Office Chair	Home Decor Ltd
	T-Shirt	Fashion World
	SQL Book	Book House
	Microwave Oven	Kitchen Solutions

Show orders handled by each employee

```

SELECT
    e.employee_name,
    o.order_id,
    o.order_date
FROM Employees e
INNER JOIN Orders o
ON e.employee_id = o.employee_id;

```

OUTPUT

	employee_name	order_id	order_date
▶	Arun	ORD001	2026-07-01
	Meena	ORD002	2026-07-02
	David	ORD003	2026-07-03
	Fatima	ORD004	2026-07-04
	Suresh	ORD005	2026-07-05

Display orders with shipper names

```
SELECT
    o.order_id,
    s.shipper_name,
    o.order_status
FROM Orders o
INNER JOIN Shippers s
ON o.shipper_id = s.shipper_id;
```

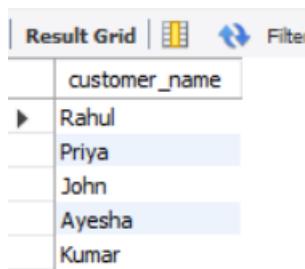
OUTPUT

Result Grid   Filter Rows:			
	order_id	shipper_name	order_status
▶	ORD001	Blue Dart	Delivered
	ORD002	DTDC	Shipped
	ORD003	Delhivery	Pending
	ORD004	India Post	Delivered
	ORD005	Ecom Express	Processing

Show customers who placed at least one order

```
SELECT DISTINCT
    c.customer_name
FROM Customers c
INNER JOIN Orders o
ON c.customer_id = o.customer_id;
```

OUTPUT



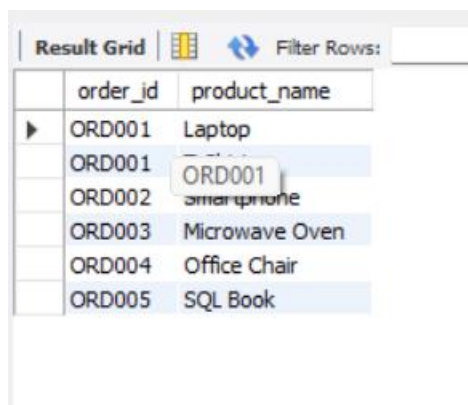
A screenshot of a database application's 'Result Grid'. The grid has a header row with 'customer_name'. Below it are five rows of data: 'Rahul', 'Priya', 'John', 'Ayesha', and 'Kumar'. The first column has a small triangle icon next to each row. The interface includes a 'Filter' button and a 'Filter Rows' input field.

customer_name
Rahul
Priya
John
Ayesha
Kumar

Display products purchased in each order

```
SELECT
    o.order_id,
    p.product_name
FROM Orders o
INNER JOIN Order_Items oi
ON o.order_id = oi.order_id
INNER JOIN Products p
ON oi.product_id = p.product_id;
```

OUTPUT



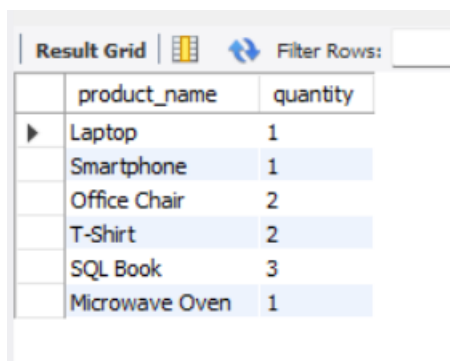
A screenshot of a database application's 'Result Grid'. The grid has two columns: 'order_id' and 'product_name'. It contains five rows of data: 'ORD001 Laptop', 'ORD001 smartphone', 'ORD002 Microwave Oven', 'ORD004 Office Chair', and 'ORD005 SQL Book'. The first column has a small triangle icon next to each row. The interface includes a 'Filter Rows' input field.

order_id	product_name
ORD001	Laptop
ORD001	smartphone
ORD002	Microwave Oven
ORD004	Office Chair
ORD005	SQL Book

Display product name and quantity ordered

```
SELECT
    p.product_name,
    oi.quantity
FROM Products p
INNER JOIN Order_Items oi
ON p.product_id = oi.product_id;
```

OUTPUT



The screenshot shows a database query result grid with the following data:

	product_name	quantity
▶	Laptop	1
	Smartphone	1
	Office Chair	2
	T-Shirt	2
	SQL Book	3
	Microwave Oven	1

Display employee names and customer names

```
SELECT
    e.employee_name,
    c.customer_name
FROM Orders o
INNER JOIN Employees e
ON o.employee_id = e.employee_id
INNER JOIN Customers c
ON o.customer_id = c.customer_id;
```

OUTPUT

	employee_name	customer_name
▶	Arun	Rahul
	Meena	Priya
	David	John
	Fatima	Ayesha
	Suresh	Kumar

Display customer, order, product, quantity and price

SELECT

c.customer_name,

o.order_id,

p.product_name,

oi.quantity,

oi.unit_price

FROM Customers c

INNER JOIN Orders o

ON c.customer_id = o.customer_id

INNER JOIN Order_Items oi

ON o.order_id = oi.order_id

INNER JOIN Products p

ON oi.product_id = p.product_id;

OUTPUT

	customer_name	order_id	product_name	quantity	unit_price
▶	Rahul	ORD001	Laptop	1	55000.00
	Rahul	ORD001	T-Shirt	2	800.00
	Priya	ORD002	Smartphone	1	25000.00
	John	ORD003	Microwave Oven	1	12000.00
	Ayesha	ORD004	Office Chair	2	4500.00
	Kumar	ORD005	SQL Book	3	650.00

Display supplier name with products sold

SELECT

s.supplier_name,

p.product_name

FROM Suppliers s

INNER JOIN Products p

ON s.supplier_id = p.supplier_id;

OUTPUT

	supplier_name	product_name
▶	ABC Electronics	Laptop
	ABC Electronics	Smartphone
	Home Decor Ltd	Office Chair
	Fashion World	T-Shirt
	Book House	SQL Book
	Kitchen Solutions	Microwave Oven

Display category-wise product count

```
SELECT
    c.category_name,
    COUNT(p.product_id) AS Total_Products
FROM Categories c
INNER JOIN Products p
ON c.category_id = p.category_id
GROUP BY c.category_name;
```

OUTPUT

	category_name	Total_Products
▶	Electronics	2
	Furniture	1
	Clothing	1
	Books	1
	Home Appliances	1

Find total orders for each customer

```
SELECT
    c.customer_name,
    COUNT(o.order_id) AS Total_Orders
FROM Customers c
LEFT JOIN Orders o
ON c.customer_id = o.customer_id
GROUP BY c.customer_name
ORDER BY Total_Orders DESC;
```

OUTPUT

Result Grid			Filter Rows:
	customer_name	Total_Orders	
▶	Rahul	1	
	Priya	1	
	John	1	
	Ayesha	1	
	Kumar	1	

Find total sales by employee

```
SELECT
    e.employee_name,
    SUM(oi.quantity * oi.unit_price) AS Total_Sales
FROM Employees e
JOIN Orders o
    ON e.employee_id = o.employee_id
JOIN Order_Items oi
    ON o.order_id = oi.order_id
GROUP BY e.employee_name
ORDER BY Total_Sales DESC;
```

OUTPUT

Find total revenue by category

```
SELECT
    c.category_name,
    SUM(oi.quantity * oi.unit_price) AS Revenue
FROM Categories c
```

```

JOIN Products p
    ON c.category_id = p.category_id
JOIN Order_Items oi
    ON p.product_id = oi.product_id
GROUP BY c.category_name
ORDER BY Revenue DESC;

```

OUTPUT

Result Grid   Filter Rows:		
	employee_name	Total_Sales
▶	Arun	56600.00
	Meena	25000.00
	David	12000.00
	Fatima	9000.00
	Suresh	1950.00

Display the most expensive product in every category

```

WITH ProductRank AS
(
    SELECT
        c.category_name,
        p.product_name,
        p.unit_price,
        DENSE_RANK() OVER
        (
            PARTITION BY c.category_name
            ORDER BY p.unit_price DESC
        ) AS RankNo
    FROM Products p
    JOIN Categories c

```

```
        ON p.category_id = c.category_id
    )
```

```
SELECT
    category_name,
    product_name,
    unit_price
FROM ProductRank
WHERE RankNo = 1;
```

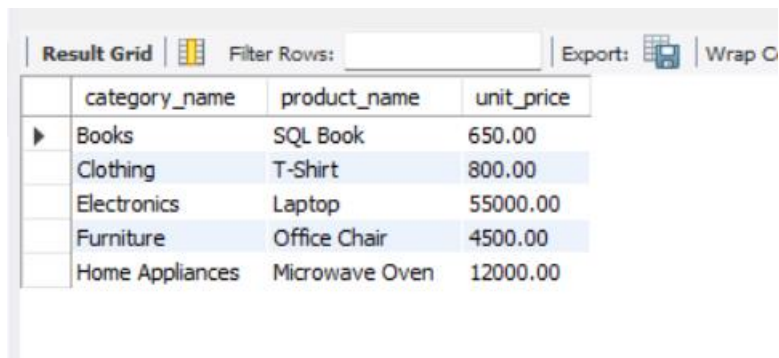
OUTPUT

Result Grid		Filter Rows:
	category_name	Revenue
▶	Electronics	80000.00
	Home Appliances	12000.00
	Furniture	9000.00
	Books	1950.00
	Clothing	1600.00

Find products never ordered

```
SELECT
    p.product_name
FROM Products p
LEFT JOIN Order_Items oi
    ON p.product_id = oi.product_id
WHERE oi.product_id IS NULL;
```

OUTPUT



The screenshot shows a database interface with a 'Result Grid' tab. Above the grid, there is a 'Filter Rows' input field and an 'Export' button with a grid icon. The grid itself has four columns: an empty column for expansion, 'category_name', 'product_name', and 'unit_price'. It contains five rows of data, with the first row expanded to show a sub-row.

	category_name	product_name	unit_price
▶	Books	SQL Book	650.00
	Clothing	T-Shirt	800.00
	Electronics	Laptop	55000.00
	Furniture	Office Chair	4500.00
	Home Appliances	Microwave Oven	12000.00

Find customers who never placed orders

SELECT

 c.customer_name

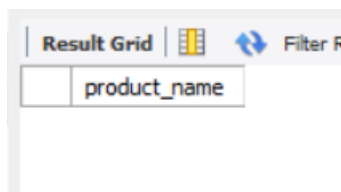
FROM Customers c

LEFT JOIN Orders o

 ON c.customer_id = o.customer_id

WHERE o.order_id IS NULL;

OUTPUT



The screenshot shows a database interface with a 'Result Grid' tab. Above the grid, there is a 'Filter Rows' input field and a 'Filter By' button with a funnel icon. The grid has one column labeled 'product_name'.

product_name

Display suppliers without products

SELECT

 s.supplier_name

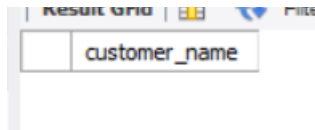
FROM Suppliers s

LEFT JOIN Products p

 ON s.supplier_id = p.supplier_id

WHERE p.product_id IS NULL;

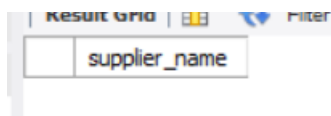
OUTPUT



Find top five customers by spending

```
SELECT
    c.customer_name,
    SUM(oi.quantity * oi.unit_price) AS TotalSpent
FROM Customers c
JOIN Orders o
    ON c.customer_id = o.customer_id
JOIN Order_Items oi
    ON o.order_id = oi.order_id
GROUP BY c.customer_name
ORDER BY TotalSpent DESC
LIMIT 5;
```

OUTPUT



Display monthly sales by employee

```
SELECT
    e.employee_name,
    MONTH(o.order_date) AS SalesMonth,
    SUM(oi.quantity * oi.unit_price) AS TotalSales
```

```

FROM Employees e
JOIN Orders o
    ON e.employee_id = o.employee_id
JOIN Order_Items oi
    ON o.order_id = oi.order_id
GROUP BY
    e.employee_name,
    MONTH(o.order_date)
ORDER BY SalesMonth;

```

OUTPUT

Result Grid			Filter Rows:
	customer_name	TotalSpent	
▶	Rahul	56600.00	
	Priya	25000.00	
	John	12000.00	
	Ayesha	9000.00	
	Kumar	1950.00	

##Find the best-selling product

```

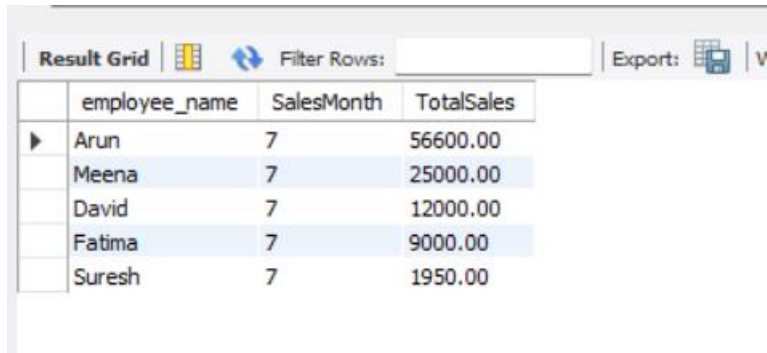
SELECT
    p.product_id,
    p.product_name,
    SUM(oi.quantity) AS Total_Quantity_Sold
FROM Products p
INNER JOIN Order_Items oi
    ON p.product_id = oi.product_id
GROUP BY
    p.product_id,
    p.product_name

```

ORDER BY Total_Quantity_Sold DESC

LIMIT 1;

OUTPUT



The screenshot shows a database interface with a 'Result Grid' tab. It includes a 'Filter Rows' input field and an 'Export' button. The table displays the following data:

	employee_name	SalesMonth	TotalSales
▶	Arun	7	56600.00
	Meena	7	25000.00
	David	7	12000.00
	Fatima	7	9000.00
	Suresh	7	1950.00

Display supplier-wise revenue

SELECT

s.supplier_name,

SUM(oi.quantity * oi.unit_price) AS Revenue

FROM Suppliers s

JOIN Products p

ON s.supplier_id = p.supplier_id

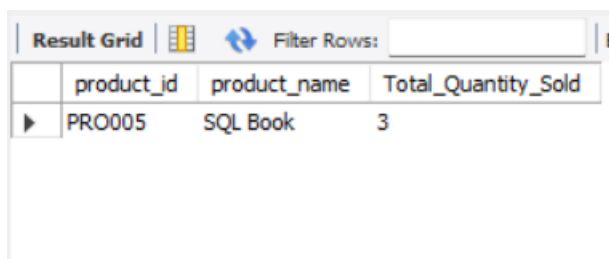
JOIN Order_Items oi

ON p.product_id = oi.product_id

GROUP BY s.supplier_name

ORDER BY Revenue DESC;

OUTPUT



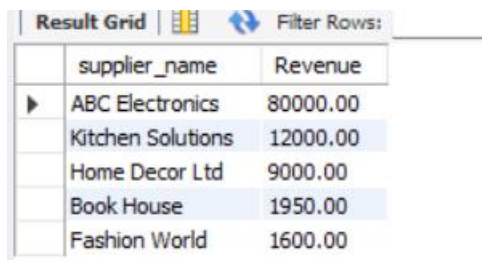
The screenshot shows a database interface with a 'Result Grid' tab. It includes a 'Filter Rows' input field and an 'Export' button. The table displays the following data:

	product_id	product_name	Total_Quantity_Sold
▶	PRO005	SQL Book	3

Find average order value per customer

```
SELECT
    c.customer_name,
    AVG(OrderValue) AS AverageOrderValue
FROM
    (
        SELECT
            o.order_id,
            o.customer_id,
            SUM(oi.quantity * oi.unit_price) AS OrderValue
        FROM Orders o
        JOIN Order_Items oi
            ON o.order_id = oi.order_id
        GROUP BY
            o.order_id,
            o.customer_id
    ) x
JOIN Customers c
    ON x.customer_id = c.customer_id
GROUP BY c.customer_name;
```

OUTPUT



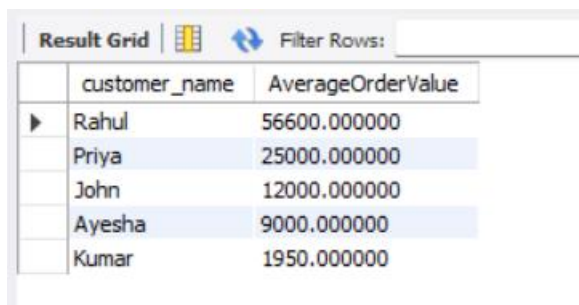
A screenshot of a database result grid. The grid has two columns: 'supplier_name' and 'Revenue'. It contains five rows of data. The first row is 'ABC Electronics' with a revenue of 80000.00. The second row is 'Kitchen Solutions' with a revenue of 12000.00. The third row is 'Home Decor Ltd' with a revenue of 9000.00. The fourth row is 'Book House' with a revenue of 1950.00. The fifth row is 'Fashion World' with a revenue of 1600.00. The grid is titled 'Result Grid' and has a 'Filter Rows' button.

	supplier_name	Revenue
▶	ABC Electronics	80000.00
	Kitchen Solutions	12000.00
	Home Decor Ltd	9000.00
	Book House	1950.00
	Fashion World	1600.00

Display orders with more than three products

```
SELECT
    order_id,
    COUNT(product_id) AS TotalProducts
FROM Order_Items
GROUP BY order_id
HAVING COUNT(product_id) > 3;
```

OUTPUT



A screenshot of a database result grid. The grid has two columns: 'customer_name' and 'AverageOrderValue'. It contains five rows of data. The first row is 'Rahul' with an average order value of 56600.000000. The second row is 'Priya' with an average order value of 25000.000000. The third row is 'John' with an average order value of 12000.000000. The fourth row is 'Ayesha' with an average order value of 9000.000000. The fifth row is 'Kumar' with an average order value of 1950.000000. The grid is titled 'Result Grid' and has a 'Filter Rows' button.

	customer_name	AverageOrderValue
▶	Rahul	56600.000000
	Priya	25000.000000
	John	12000.000000
	Ayesha	9000.000000
	Kumar	1950.000000

Find customers ordering from multiple categories

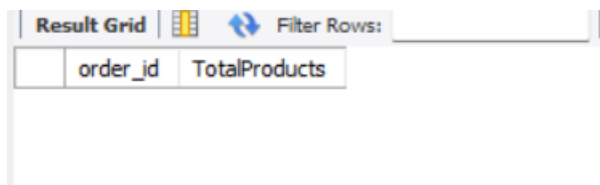
```
SELECT
    c.customer_name,
    COUNT(DISTINCT p.category_id) AS CategoriesPurchased
```

```

FROM Customers c
JOIN Orders o
    ON c.customer_id = o.customer_id
JOIN Order_Items oi
    ON o.order_id = oi.order_id
JOIN Products p
    ON oi.product_id = p.product_id
GROUP BY c.customer_name
HAVING COUNT(DISTINCT p.category_id) > 1;

```

OUTPUT



order_id	TotalProducts
----------	---------------

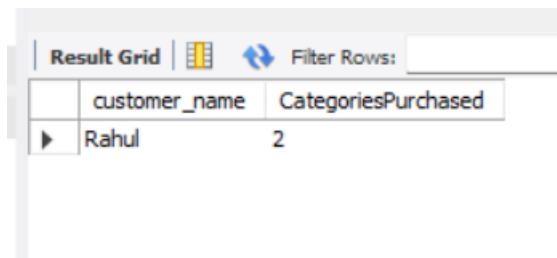
Display categories with no sales

```

SELECT
    c.category_name
FROM Categories c
LEFT JOIN Products p
    ON c.category_id = p.category_id
LEFT JOIN Order_Items oi
    ON p.product_id = oi.product_id
WHERE oi.order_item_id IS NULL;

```

OUTPUT

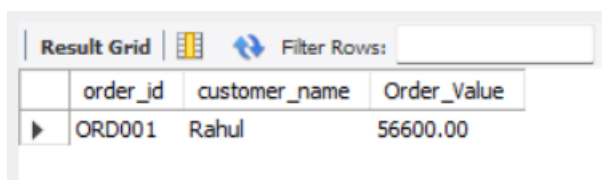


	customer_name	CategoriesPurchased
▶	Rahul	2

Display the highest-value order

```
SELECT
    o.order_id,
    c.customer_name,
    SUM(oi.quantity * oi.unit_price) AS Order_Value
FROM Orders o
INNER JOIN Customers c
    ON o.customer_id = c.customer_id
INNER JOIN Order_Items oi
    ON o.order_id = oi.order_id
GROUP BY
    o.order_id,
    c.customer_name
ORDER BY Order_Value DESC
LIMIT 1;
```

OUTPUT



	order_id	customer_name	Order_Value
▶	ORD001	Rahul	56600.00

Generate a complete invoice report using all tables

```
SELECT
    o.order_id,
    o.order_date,
    c.customer_name,
    e.employee_name,
    s.shipper_name,
    p.product_name,
    cat.category_name,
    sup.supplier_name,
    oi.quantity,
    oi.unit_price,
    (oi.quantity * oi.unit_price) AS LineTotal,
    o.order_status
FROM Orders o
JOIN Customers c
    ON o.customer_id = c.customer_id
JOIN Employees e
    ON o.employee_id = e.employee_id
JOIN Shippers s
    ON o.shipper_id = s.shipper_id
JOIN Order_Items oi
    ON o.order_id = oi.order_id
JOIN Products p
    ON oi.product_id = p.product_id
JOIN Categories cat
    ON p.category_id = cat.category_id
JOIN Suppliers sup
```

ON p.supplier_id = sup.supplier_id

ORDER BY

o.order_id,

p.product_name;

OUTPUT

Result Grid

Filter Rows:

Export:

Wrap Cell Contents:

	order_id	order_date	customer_name	employee_name	shipper_name	product_name	category_name	supplier_name	quantity	unit_price	LineTotal	order_status
▶	ORD001	2026-07-01	Rahul	Arun	Blue Dart	Laptop	Electronics	ABC Electronics	1	55000.00	55000.00	Delivered
	ORD001	2026-07-01	Rahul	Arun	Blue Dart	T-Shirt	Clothing	Fashion World	2	800.00	1600.00	Delivered
	ORD002	2026-07-02	Priya	Meena	DTDC	Smartphone	Electronics	ABC Electronics	1	25000.00	25000.00	Shipped
	ORD003	2026-07-03	John	David	Delhivery	Microwave Oven	Home Appliances	Kitchen Solutions	1	12000.00	12000.00	Pending
	ORD004	2026-07-04	Ayesha	Fatima	India Post	Office Chair	Furniture	Home Decor Ltd	2	4500.00	9000.00	Delivered
	ORD005	2026-07-05	Kumar	Suresh	Ecom Express	SQL Book	Books	Book House	3	650.00	1950.00	Processing